

1)

$$a(kT) = (-1)^{|k|} + (0.25)^{|k|}$$

$$(-1)^{|k|} = e^{j2\pi kT \frac{1}{2T}} \longleftrightarrow \delta_{\frac{1}{2T}}\left(f - \frac{1}{2T}\right)$$

$$T \cdot \sum_{k=-\infty}^{+\infty} (0.25)^{|k|} e^{-j2\pi f kT} =$$

$$= T \left[\sum_{k=0}^{+\infty} 0.25^k e^{-j2\pi f kT} + \sum_{k=0}^{+\infty} (0.25)^k e^{+j2\pi f kT} - 1 \right]$$

$$= \frac{T}{1 - 0.25 e^{-j2\pi f T}} + \frac{T}{1 - 0.25 e^{+j2\pi f T}} - T$$

$$= T \frac{1 - (0.25)^2}{1 - 0.5 \cos 2\pi f T + (0.25)^2}$$

$$A(f) = \delta_{\frac{1}{2T}}\left(f - \frac{1}{2T}\right) + \frac{T(1 - (0.25)^2)}{1 - 0.5 \cos 2\pi f T + (0.25)^2}$$

2)

$$a(kT) = \frac{1}{T} (0.25)^{|k|}$$


$$X(f) = A(f) \cdot \mathcal{F}\left[\text{sinc}\left(\frac{2t}{T}\right)\right]$$

↑
produces

↑
bande passante $B = \frac{1}{T}$

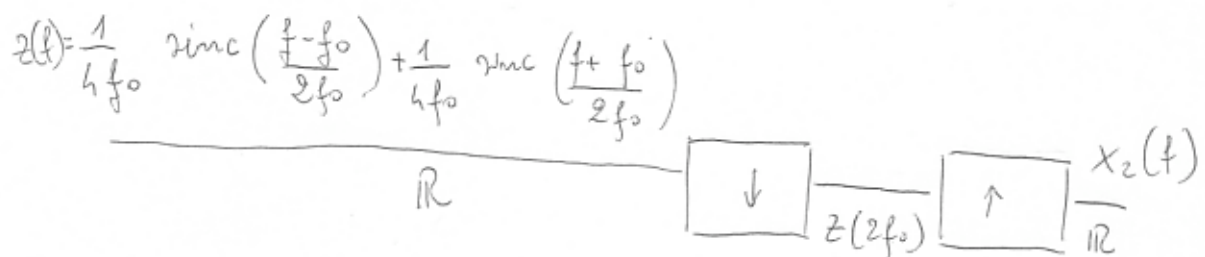
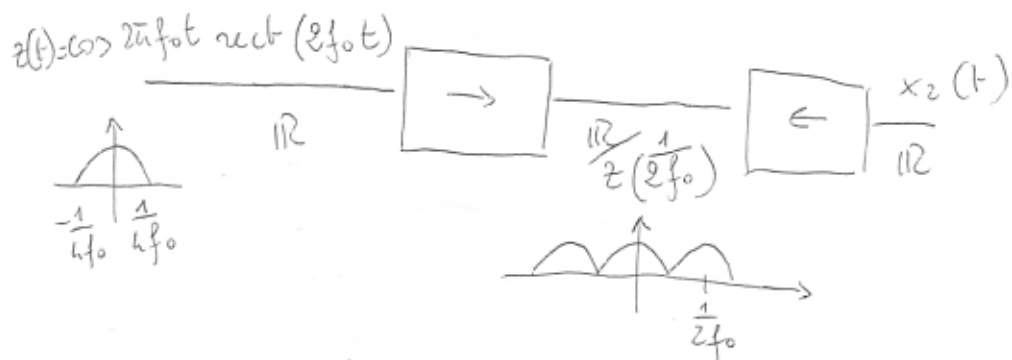
$$F_c = 2B = \frac{2}{T}$$

$$3) x(t) = |(1 + \nu \cos^2 t) \cos 2\pi f_0 t| = (1 + \nu \cos^2 t) |\cos 2\pi f_0 t|$$

$$= x_1(t) \cdot x_2(t)$$

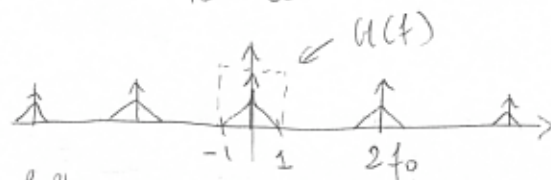
$$X(f) = X_1 * X_2(f)$$

$$X_1(f) = \delta_R(f) + \text{triangle}(f)$$



$$X_2(f) = 2f_0 \cdot \sum_{k=-\infty}^{+\infty} z(2kf_0) \delta_R(f - 2kf_0)$$

$$X(f) = 2f_0 \sum_{k=-\infty}^{+\infty} z(2kf_0) X_1(f - 2kf_0)$$



Dopo il filtraggio con $H(f) = \text{rect}\left(\frac{f}{2}\right)$ resta $Y(f) = 2f_0 z(0) X_1(f)$

$$y(t) = \frac{2}{\pi} (1 + \cos^2 t)$$

4)

$$\begin{aligned}
 E_x &= \int_{-\infty}^{+\infty} |2j e^{-4(t-1)^2} e^{2j}|^2 dt \\
 &= 4 \int_{-\infty}^{+\infty} e^{-8(t-1)^2} dt = 4 \int_{-\infty}^{+\infty} e^{-8t^2} dt \\
 &= 4 \int_{-\infty}^{+\infty} e^{-\pi \left(\frac{\sqrt{8}}{\sqrt{\pi}} t\right)^2} dt = 4 \sqrt{\frac{\pi}{8}} \cdot e^{-\pi \left(\frac{1}{\sqrt{8}}\right)^2} \Big|_{f=0} \\
 &= \sqrt{2\pi}
 \end{aligned}$$

5)

$$B(f) = \frac{A(f)}{1 - 0.5 e^{-j2\pi f 2T}}$$

è tratto da un filtro con funzione di f .

$$H_z(z) = \frac{1}{1 - 0.5 z^{-2}} = \frac{1/2}{1 - \frac{1}{\sqrt{2}} z^{-1}} + \frac{1/2}{1 + \frac{1}{\sqrt{2}} z^{-1}}$$

i poli $p_{1,2} = \pm \frac{1}{\sqrt{2}}$ sono in modulo $< 1 \Rightarrow$ il filtro è stabile in senso BIBO

$$h(nT) = \frac{1}{2T} \left(\frac{1}{\sqrt{2}}\right)^n 1_0(nT) + \frac{1}{2T} \left(-\frac{1}{\sqrt{2}}\right)^n 1_0(nT)$$

$$h(nT) = |H(f_0)| \cos\left(2\pi \frac{nT}{4T} + \angle H(f_0)\right); \quad f_0 = \frac{1}{4T}$$

$$H\left(\frac{1}{4T}\right) = \frac{2}{3}$$

$$h(nT) = \frac{2}{3} \cos 2\pi \frac{nT}{4}$$